

CSCI 6502

Big Data Analytics: Systems, Algorithms, and Applications

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Machine Learning Engineer (97.0 / 100)	82.8 / 100

83	100	+38
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

This course studies state-of-the-art practice and research on efficient and effective systems and algorithms design for managing and exploring massive amounts of digital data in various application domains. The course takes an integrated approach that studies all three aspects of big data analytics: systems, algorithms, and applications. Specifically, this course covers big data systems for MapReduce, NoSQL, stream processing, deep learning, mobile/wearable/IoT sensing, as well as practical use of indexing, sketching, recommendation, graph, and deep learning algorithms. Domain-specific data management and analysis, such as those in online social networks, scientific discovery, business intelligence, health informatics, urban computing, are also covered.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	97.0	100.0	+3.0	LOW
AI Engineer	94.8	100.0	+5.2	LOW
ML Engineer	93.4	100.0	+6.6	LOW
Applied Scientist	85.9	100.0	+14.1	LOW
Data Scientist	64.2	100.0	+35.8	HIGH
MLOps Engineer	61.6	100.0	+38.4	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Model Deployment & Serving 2 of 6 roles

production

LLMs & Generative AI 1 of 6 roles

storage

Feature Engineering 1 of 6 roles

preprocessing

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **MEDIUM** **Consider Model Deployment & Serving content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: production.

2 **LOW** **Optional: LLMs & Generative AI** — would broaden appeal to one additional role. Candidate terms: storage.

3 **LOW** **Optional: Feature Engineering** — would broaden appeal to one additional role. Candidate terms: preprocessing.

GAP KEYWORDS BY ROLE

AI Engineer

94.8 → 100.0 (+5.2 pts)

robotics engineering models infrastructure build software technical problems

ML Engineer

93.4 → 100.0 (+6.6 pts)

media machine models production engineering starting infrastructure world

Applied Scientist

85.9 → 100.0 (+14.1 pts)

lab storage literature problems models technical industry engineering

Data Scientist

64.2 → 100.0 (+35.8 pts)

toolkit sets product characteristics evolution analytical build engineering

MLOps Engineer

61.6 → 100.0 (+38.4 pts)

infrastructure detection production machine preprocessing media image engineering

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.